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Inventor(s)	Griffin; Leigh et al.

Quantum authentication of protected resources

Abstract

An access request to allow a requestor process to access a protected resource is received. A qubit authentication state that identifies a desired state of a qubit is sent to the requestor process. It is determined that the requestor process has caused the qubit to have the qubit authentication state. Based at least in part on determining that the requestor process has caused the qubit to have the qubit authentication state, the requestor process is granted access to the protected resource.

Inventors: **Griffin; Leigh (Waterford, IE), Coady; Stephen (Waterford, IE)**

Applicant: **Red Hat, Inc.** (Raleigh, NC)

Family ID: **87574271**

Assignee: **Red Hat, Inc. (Raleigh, NC)**

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Primary Examiner: Moise; Emmanuel L

Assistant Examiner: Yu; Xiang

Attorney, Agent or Firm: Dority & Manning, P.A.

Background/Summary

BACKGROUND
(1) As quantum computing becomes more prevalent, it is increasingly important to ensure that quantum computing environments are secure.
SUMMARY
(2) The examples disclosed herein implement quantum authentication of protected resources.
(3) In one example a method is provided. The method includes receiving, by a computing device including a processor device, an access request to allow a requestor process to access a protected resource. The method further includes sending, by the computing device to the requestor process, a qubit authentication state that identifies a desired state of a qubit. The method further includes determining, by the computing device, that the requestor process has caused the qubit to have the qubit authentication state. The method further includes, based at least in part on determining that the requestor process has caused the qubit to have the qubit authentication state, granting, by the computing device to the requestor process, access to the protected resource.
(4) In another example another method is provided. The method includes receiving, by a computing device including a processor device, an access request to allow a requestor process to access a protected resource. The method further includes sending, by the computing device to a first quantum computing system, a request for a quantum authentication. The method further includes

receiving, by the computing device from the first quantum computing system, a qubit authentication state that identifies a desired state of a qubit. The method further includes sending, by the computing device to the requestor process, the qubit authentication state. The method further includes receiving, by the computing device from the first quantum computing system, a successful quantum authentication message indicating a successful quantum authentication by the requestor process. The method further includes, based at least in part on the successful quantum authentication message, granting, by the computing device to the requestor process, access to the protected resource.

(5) In another example a computing device is provided. The computing device includes a memory, and a processor device coupled to the memory. The processor device is to receive an access request to allow a requestor process to access a protected resource. The processor device is further to send, to the requestor process, a qubit authentication state that identifies a desired state of a qubit. The processor device is further to determine that the requestor process has caused the qubit to have the qubit authentication state. The processor device is further to, based at least in part on determining that the requestor process has caused the qubit to have the qubit authentication state, grant, to the requestor process, access to the protected resource.

(6) Individuals will appreciate the scope of the disclosure and realize additional aspects thereof after reading the following detailed description of the examples in association with the accompanying drawing figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawing figures incorporated in and forming a part of this specification illustrate several aspects of the disclosure and, together with the description, serve to explain the principles of the disclosure.

(2) FIG. **1** is a block diagram of an environment suitable for implementing quantum authentication of protected resources according to one implementation;

(3) FIG. **2** is a flowchart of a method for quantum authentication of protected resources according to one implementation;

(4) FIG. **3** is a block diagram of an environment suitable for implementing quantum authentication of protected resources according to another implementation;

(5) FIG. **4** is a flowchart of a method for quantum authentication of protected resources according to another implementation;

(6) FIGS. **5**A-**5**B illustrate a sequence diagram illustrating actions taken and messages communicated by various components illustrated in FIG. **3** to implement quantum authentication of protected resources according to one implementation;

(7) FIG. **6** is a block diagram of an environment suitable for implementing quantum authentication of protected resources according to yet another implementation;

(8) FIG. **7** is a simplified block diagram of the environment illustrated in FIG. **1** according to one implementation; and

(9) FIG. **8** is a block diagram of a computing device suitable for implementing the implementations disclosed herein.

DETAILED DESCRIPTION

(10) The examples set forth below represent the information to enable individuals to practice the examples and illustrate the best mode of practicing the examples. Upon reading the following description in light of the accompanying drawing figures, individuals will understand the concepts of the disclosure and will recognize applications of these concepts not particularly addressed herein. It should be understood that these concepts and applications fall within the scope of the

disclosure and the accompanying claims.
(11) Any flowcharts discussed herein are necessarily discussed in some sequence for purposes of illustration, but unless otherwise explicitly indicated, the examples are not limited to any particular sequence of steps. The use herein of ordinals in conjunction with an element is solely for distinguishing what might otherwise be similar or identical labels, such as “first message” and “second message,” and does not imply a priority, a type, an importance, or other attribute, unless otherwise stated herein. The term “about” used herein in conjunction with a numeric value means any value that is within a range of ten percent greater than or ten percent less than the numeric value. As used herein and in the claims, the articles “a” and “an” in reference to an element refers to “one or more” of the element unless otherwise explicitly specified. The word “or” as used herein and in the claims is inclusive unless contextually impossible. As an example, the recitation of A or B means A, or B, or both A and B.
(12) As quantum computing becomes more prevalent, it is increasingly likely that hacking and other nefarious activities will focus on quantum computing environments. A quantum computing system can typically operate in both a classical computing mode, where data is represented in the binary values of 0 and 1, and in a quantum computing mode, where computations utilize quantum-mechanical phenomena, such as superposition and entanglement. A nefarious process that enters a quantum computing system may have been designed for classical computing device environments but ultimately spread to the quantum computing system.
(13) The examples disclosed herein implement quantum authentication of protected resources. In particular, a process that requests access to a protected resource, such as an online service, a web page, a file, a database, or the like, must establish that the process has the capability of manipulating a qubit in a specified manner in order to be granted access to the protected resource. Nefarious programs that do not have the capability of manipulating a qubit are thus prevented from accessing a protected resource.
(14) FIG. **1** is a block diagram of an environment **10** suitable for implementing quantum authentication of protected resources according to one example. The environment **10** includes a computing device, such as a quantum computing system **12-1** which includes a memory **14-1**, a processor device **16-1**, and implements a plurality of qubits **18-1-18**-N. The quantum computing system **12-1** operates in a quantum environment but is capable of operating using classical computing principles or quantum computing principles. When using quantum computing principles, the quantum computing system **12-1** performs computations that utilize quantum-mechanical phenomena, such as superposition and/or entanglement states. The quantum computing system **12-1** may operate under certain environmental conditions, such as at or near zero degrees (0°) Kelvin. When using classical computing principles, the quantum computing system **12-1** utilizes binary digits that have a value of either zero (0) or one (1).
(15) The quantum computing system **12-1** includes a multi-factor authentication (MFA) service **28-1** that is configured to ensure that a process that requests access to a protected resource **30**, in this example a file **32-1** stored on a storage device **33**, is properly authenticated prior to being granted permission to access the file **32-1**. The term “access” in the context of the file **32-1** may mean, for example, reading the file **32-1**, modifying the file **32-1**, copying the file **32-1**, deleting the file **32-1**, or any other manipulation or action associated with the file **32-1**. The examples disclosed herein may operate with any type of protected resource **30** for which protection may be desirable, such as, by way of non-limiting example, a database **32-2**, an online service **32-3**, or an application programming interface (API) endpoint **32-4**, and the term “access” may differ depending on the particular protected resource **30**. For example, with regard to the database **32-2**, the MFA service **28-1** may prohibit reads, writes, modifications, or deletes unless first authenticated by the MFA service **28-1**. With regard to the online service **32-3**, the MFA service **28-1** may prohibit logging into the online service **32-3** unless first authenticated by the MFA service **28-1**. With regard to the API endpoint **32-4**, the MFA service **28-1** may prohibit invoking the API endpoint **32-4** unless first

authenticated by the MFA service **28-1**.
(16) In this example, a requestor process **34** attempts to access the file **32-1**. The word "requestor" in the term "requestor process" is used solely to refer to the action of the process, and any process that desires access to a resource can be a requestor process. As part of the attempt to access the file **32-1**, the MFA service **28-1** is invoked. The MFA service **28-1** may initially send the requestor process **34** a request for a classical authentication, such as a request for a user identifier and a password, or any other desirable authentication mechanism used on classical computing devices to authenticate a process. In response, the requestor process **34** sends authentication information to the MFA service **28-1**. The MFA service **28-1**, in this example, determines that the authentication information is valid.
(17) The MFA service **28-1** then initiates a quantum authentication. The MFA service **28-1** generates a qubit authentication state. The qubit authentication state is a state of a qubit into which the requestor process **34** will be asked to place a qubit. The qubit authentication state may comprise a particular spin state, polarization state, or any other attribute or state of a qubit. In some implementations, the qubit authentication state may comprise a textual identifier that the requestor process is to store in the qubit. In some examples, the MFA service **28-1** may access qubit authentication states **36** to determine the qubit authentication state. The qubit authentication states **36** may comprise a plurality of entries, each of which identifies a different qubit authentication state, such as a different combination of spin state, polarization state, and/or other physical characteristic of qubit. The MFA service **28-1** may randomly select an entry in the qubit authentication states **36** to determine the qubit authentication state.
(18) The MFA service **28-1** may then reserve an available qubit, such as the qubit **18-1**. The MFA service **28-1** provides, to the requestor process **34**, a qubit identifier that identifies the qubit **18-1**, and the qubit authentication state. The requestor process **34** receives the qubit identifier that identifies the qubit **18-1**, and the qubit authentication state, and performs the desired manipulations on the qubit **18-1** to place the qubit **18-1** into the qubit authentication state. The requestor process **34** may inform the MFA service **28-1** that the requestor process **34** has caused the qubit **18-1** to have the qubit authentication state, or, alternatively, the MFA service **28-1** may monitor the qubit **18-1** to determine that the qubit **18-1** has been modified.
(19) The MFA service **28-1** determines the physical characteristics of the qubit **18-1** and compares the physical characteristics to the qubit authentication state previously sent to the requestor process **34**. In this example, the MFA service **28-1** determines that the physical characteristics match the qubit authentication state. In response to this determination, the MFA service **28-1** grants the requestor process **34** access to the file **32-1**. If the MFA service **28-1** had determined that the physical characteristics did not match the qubit authentication state, the MFA service **28-1** would deny the requestor process **34** access to the protected resource **30**.
(20) In some examples, after providing the qubit identifier that identifies the qubit **18-1** and the qubit authentication state to the requestor process **34**, the MFA service **28-1** may set a timer for a determined period of time, such as 5 seconds, 10 seconds, or any other suitable interval. If the requestor process **34** does not manipulate the qubit **18-1** by the time the timer expires, the MFA service **28-1** denies access to the protected resource **30** by the requestor process **34**.
(21) It is noted that, because the MFA service **28-1** is a component of the quantum computing system **12-1**, functionality implemented by the MFA service **28-1** may be attributed to the quantum computing system **12-1** generally. Moreover, in examples where the MFA service **28-1** comprises software instructions that program the processor device **16-1** to carry out functionality discussed herein, functionality implemented by the MFA service **28-1** may be attributed herein to the processor device **16-1**.
(22) FIG. **2** is a flowchart of a method for quantum authentication of protected resources according to one implementation. FIG. **2** will be discussed in conjunction with FIG. **1**. The quantum computing system **12-1** receives, from the requestor process **34**, an access request to allow the

requestor process **34** to access a protected resource **30** (FIG. **2**, block **1000**). The quantum computing system **12-1** sends, to the requestor process **34**, a qubit authentication state that identifies a desired state of a qubit (FIG. **2**, block **1002**). The quantum computing system **12-1** determines that the requestor process **34** has caused the qubit **18-1** to have the qubit authentication state (FIG. **2**, block **1004**). Based at least in part on determining that the requestor process **34** has caused the qubit **18-1** to have the qubit authentication state, the quantum computing system **12-1** grants the requestor process **34** access to the protected resource **30** (FIG. **2**, block **1006**).

(23) FIG. **3** is a block diagram of an environment **10-1** suitable for implementing quantum authentication of protected resources according to another example. The environment **10-1** includes the quantum computing system **12-1** and a quantum computing system **12-2**. The quantum computing system **12-2** includes a memory **14-2**, a processor device **16-2**, and implements a plurality of qubits **20-1-20-**M.

(24) The quantum computing system **12-2** operates in a quantum environment, but is capable of operating using classical computing principles or quantum computing principles. When using quantum computing principles, the quantum computing system **12-2** performs computations that utilize quantum-mechanical phenomena, such as superposition and/or entanglement states. The quantum computing system **12-2** may operate under certain environmental conditions, such as at or near zero degrees (0°) Kelvin. When using classical computing principles, the quantum computing system **12-2** utilizes binary digits that have a value of either zero (0) or one (1).

(25) The quantum computing systems **12-1** and **12-2** are communicatively coupled to one another via a classical communications channel **22** and via a quantum communications channel **24** over which physical properties of qubits may be communicated via, for example, quantum teleportation. The quantum computing systems **12-1** and **12-2** may be in physical proximity to one another or may be separated by a substantial distance.

(26) In this example, the quantum computing system **12-1** includes a teleportation service **26-1** and the quantum computing system **12-2** includes a teleportation service **26-2** (generally, teleportation services **26**). The teleportation services **26** are configured to, upon request, cause a qubit **18** implemented by the quantum computing system **12-1** to become entangled with a qubit **20** implemented by the quantum computing system **12-2**. The teleportation services **26** are also configured to, upon request, cause the physical attributes of a qubit **18** to be teleported from the quantum computing system **12-1** to the quantum computing system **12-2** using the entangled pair of qubits **18**, **20** via, for example, bell states. In this example, the teleportation services **26-1** and **26-2** have caused the qubit **18-**N and the qubit **20-**M to become entangled.

(27) In this example, the requestor process **34** attempts to access the file **32-1**. As part of the attempt to access the file **32-1**, the MFA service **28-1** is invoked. The MFA service **28-1** may initially send the requestor process **34** the request for the classical authentication, such as a user identifier and a password, or any other desirable authentication mechanism used on classical computing devices to authenticate a process. In response, the requestor process **34** sends authentication information to the MFA service **28-1**. The MFA service **28-1**, in this example, determines that the authentication information is valid. The MFA service **28-1** then initiates a quantum authentication. In particular, the MFA service **28-1** sends a request for quantum authentication to an MFA service **28-2** on the quantum computing system **12-2**.

(28) The MFA service **28-2** receives the request for quantum authentication and generates a qubit authentication state. In some examples, the MFA service **28-2** may access the qubit authentication states **36** to determine the qubit authentication state. The MFA service **28-2** sends the qubit authentication state to the MFA service **28-1**. The MFA service **28-1** receives the qubit authentication state that identifies the desired state of the qubit. The MFA service **28-1** may then reserve an available qubit, such as the qubit **18-1**. The MFA service **28-1** provides, to the requestor process **34**, a qubit identifier that identifies the qubit **18-1**, and the qubit authentication state. The requestor process **34** receives the qubit identifier that identifies the qubit **18-1**, and the qubit

authentication state, and performs the desired manipulations on the qubit **18-1** to place the qubit **18-1** in the qubit authentication state. The requestor process **34** may inform the MFA service **28-1** that the requestor process **34** has caused the qubit **18-1** to have the qubit authentication state.

(29) The MFA service **28-1** invokes the teleportation service **26-1** identifying the qubit **18-1**. The teleportation service **26-1**, using the quantum communications channel **24** and the entangled qubit **18**-N, teleports the physical characteristics of the qubit **18-1** to the qubit **20**-M. The MFA service **28-2** determines that the qubit **20**-M has been altered. The MFA service **28-2** determines the physical characteristics of the qubit **20**-M and compares the physical characteristics to the qubit authentication state previously sent to the MFA service **28-1**. In this example, the MFA service **28-2** determines that the physical characteristics match the qubit authentication state. In response to this determination, the MFA service **28-2** sends a successful quantum authentication message to the MFA service **28-1**. The MFA service **28-1** receives the successful quantum authentication message and grants the requestor process **34** access to the file **32-1**.

(30) It is noted that, because the MFA service **28-2** is a component of the quantum computing system **12-2**, functionality implemented by the MFA service **28-2** may be attributed to the quantum computing system **12-2** generally. Moreover, in examples where the MFA service **28-2** comprises software instructions that program the processor device **16-2** to carry out functionality discussed herein, functionality implemented by the MFA service **28-2** may be attributed herein to the processor device **16-2**.

(31) FIG. **4** is a flowchart of a method for quantum authentication of protected resources according to one implementation. FIG. **4** will be discussed in conjunction with FIG. **3**. The quantum computing system **12-1** receives, from the requestor process **34**, an access request to allow the requestor process **34** to access a protected resource **30** (FIG. **4**, block **2000**). The quantum computing system **12-1** sends, to the quantum computing system **12-2**, a request for a quantum authentication (FIG. **4**, block **2002**). The quantum computing system **12-1** receives, from the quantum computing system **12-2**, a qubit authentication state that identifies a desired state of a qubit (FIG. **4**, block **2004**). The quantum computing system **12-1** sends, to the requestor process **34**, the qubit authentication state (FIG. **4**, block **2006**). The quantum computing system **12-1** receives, from the quantum computing system **12-2**, a successful quantum authentication message indicating a successful quantum authentication by the requestor process **34** (FIG. **4**, block **2008**). The quantum computing system **12-1**, based at least in part on the successful quantum authentication message, grants, to the requestor process **34**, access to the protected resource (FIG. **4**, block **2010**).

(32) FIGS. **5**A-**5**B illustrate a sequence diagram illustrating actions taken and messages communicated by various components illustrated in FIG. **3** to implement quantum authentication of protected resources according to one implementation. Referring first to FIG. **5**A, initially, the teleportation service **26-1** unilaterally, or at the request of the MFA service **28-1**, reserves the qubit **18**-N for purposes of being entangled with a qubit of the quantum computing system **12-2** (FIG. **5**A, step **3000**). The teleportation service **26-1** sends a message to the teleportation service **26-2** of the quantum computing system **12-2** informing the teleportation service **26-2** that the qubit **18**-N has been reserved, and providing addressing and other suitable information to the teleportation service **26-2** to facilitate entanglement of the qubit **18**-N with the qubit **20**-M of the quantum computing system **12-2** (FIG. **5**A, step **3002**). The teleportation service **26-2** reserves the qubit **20**-M for purposes of being entangled with the qubit **18**-N, and causes an entanglement between the qubits **18**-N and **20**-M (FIG. **5**A, steps **3004**, **3006**).

(33) Subsequently, the requestor process **34** attempts to access a protected resource **30**, such as the file **32-1** (FIG. **3**) (FIG. **5**A, step **3008**). As part of the access attempt, the file system of the quantum computing system **12-1** invokes the MFA service **28-1**. The MFA service **28-1** requests a classical authentication of the requestor process **34** (FIG. **5**A, step **3010**). The requestor process **34** provides a user identifier and a password to the MFA service **28-1** (FIG. **5**A, step **3012**). The MFA service **28-1** determines that the provided user identifier and password are correct and that the

requestor process **34** has thus successfully passed the classical authentication (FIG. **5**A, step **3014**).

(34) The MFA service **28-1** sends a request to the MFA service **28-2** for a quantum authentication (FIG. **5**A, step **3016**). The MFA service **28-2** generates a qubit authentication state, and sends qubit authentication state to the MFA service **28-1** (FIG. **5**A, steps **3018-3020**). The MFA service **28-1** reserves the qubit **18-1**, and sends the qubit authentication state and a qubit identifier of the qubit **18-1** to the requestor process **34** (FIG. **5**A, steps **3022**).

(35) Referring now to FIG. **5**B, the requestor process **34** receives the qubit authentication state and the qubit identifier of the qubit **18-1**, and manipulates the qubit **18-1**, such as by performing the appropriate qubit operations, to place the qubit **18-1** into the qubit authentication state (FIG. **5**B, step **3024**). The MFA service **28-1** determines that the qubit **18-1** has been manipulated (FIG. **5**B, step **3026**). In some examples, the MFA service **28-1** may be informed by the requestor process **34** that the qubit **18-1** has been manipulated. In other examples, the MFA service **28-1** may continuously monitor the qubit **18-1** and determine when the physical state of the qubit **18-1** has changed, indicating that the qubit **18-1** has been manipulated.

(36) The MFA service **28-1** requests that the teleportation service **26-1** teleport the physical characteristics of the qubit **18-1** to the quantum computing system **12-2** (FIG. **5**B, step **3028**). The teleportation service **26-1** causes the physical characteristics of the qubit **18-1** to be teleported to the quantum computing system **12-2** using the entangled pair of qubits **18-**N and **20-**M (FIG. **5**B, step **3030**). The MFA service **28-2** determines that the qubit **20-**M has been manipulated to match the qubit authentication state (FIG. **5**B, step **3032**). The MFA service **28-2** sends a message to the MFA service **28-1** indicating that the requestor process **34** has been properly authenticated (FIG. **5**B, step **3034**). The MFA service **28-1** grants permission to the requestor process **34** to access the protected resource **30** (FIG. **5**B, step **3036**).

(37) FIG. **6** is a block diagram of an environment **10-2** suitable for implementing quantum authentication of protected resources according to yet another implementation. In this example, the MFA service **28-1** executes on a computing device **38** that is a classical computing device, and the requestor process **34** executes on a computing device **40** that is also a classical computing device.

(38) The requestor process **34** attempts to access a protected resource **30**, such as the file **32-1**. As part of the attempt to access the file **32-1**, the MFA service **28-1** is invoked. The MFA service **28-1** may initially send the requestor process **34** the request for the classical authentication, such as a user identifier and a password, or any other desirable mechanism used on classical computing devices to authenticate a process. In response, the requestor process **34** sends authentication information to the MFA service **28-1**. The MFA service **28-1**, in this example, determines that the authentication information is valid.

(39) The MFA service **28-1** then sends a message requesting a quantum authentication to an MFA service **28-3** executing on the quantum computing system **12-1**. The MFA service **28-3** sends a request for quantum authentication to the MFA service **28-2** on the quantum computing system **12-2**.

(40) The MFA service **28-2** receives the request for quantum authentication and generates a qubit authentication state. The MFA service **28-2** sends the qubit authentication state to the MFA service **28-3**. The MFA service **28-3** receives the qubit authentication state that identifies the desired state of the qubit. The MFA service **28-3** may then reserve an available qubit, such as the qubit **18-1**. The MFA service **28-3** provides, to the MFA service **28-1**, a qubit identifier that identifies the qubit **18-1**, and the qubit authentication state. The MFA service **28-1** provides, to the requestor process **34**, the qubit identifier and the qubit authentication state. The requestor process **34** receives the qubit identifier that identifies the qubit **18-1**, and the qubit authentication state, and sends commands to the quantum computing system **12-1** that cause the desired manipulations on the qubit **18-1** to place the qubit **18-1** in the qubit authentication state. The requestor process **34** may inform the MFA service **28-1** that the requestor process **34** has caused the qubit **18-1** to have the qubit authentication state. The MFA service **28-1** may then inform the MFA service **28-3** that the requestor process **34**

has indicated that the requestor process **34** has manipulated the qubit **18-1**. In other implementations, the MFA service **28-3** may detect that the qubit **18-1** has been manipulated.
(41) The MFA service **28-3** invokes the teleportation service **26-1** identifying the qubit **18-1**. The teleportation service **26-1**, using the quantum channel **24** and the entangled qubit **18**-N, teleports the physical characteristics of the qubit **18-1** to the qubit **20**-M. The MFA service **28-2** determines that the qubit **20**-M has been altered. The MFA service **28-2** determines the physical characteristics of the qubit **20**-M and compares the physical characteristics to the qubit authentication state previously sent to the MFA service **28-3**. In this example, the MFA service **28-2** determines that the physical characteristics match the qubit authentication state. In response to this determination, the MFA service **28-2** sends a successful quantum authentication message to the MFA service **28-3**. The MFA service **28-3** receives the successful quantum authentication message and sends the successful quantum authentication message to the MFA service **28-1**. The MFA service **28-1** receives the successful quantum authentication message and grants the requestor process **34** access to the file **32-1**.
(42) FIG. **7** is a simplified block diagram of the environment **10** illustrated in FIG. **1** according to one implementation. The environment **10** includes a computing device, in the form of the quantum computing system **12-1**, which in turn includes the processor device **16-1** coupled to the memory **14-1**. The processor device **16-1** is to receive an access request to allow the requestor process **34** to access a protected resource **30** such as the file **32-1**. The processor device **16-1** is further to send, to the requestor process **34**, a qubit authentication state that identifies a desired state of the qubit **18-1**. The processor device **16-1** is to determine that the requestor process **34** has caused the qubit **18-1** to have the qubit authentication state. The processor device **16-1** is to, based at least in part on determining that the requestor process **34** has caused the qubit **18-1** to have the qubit authentication state, grant, to the requestor process **34**, access to the protected resource **30**.
(43) FIG. **8** is a block diagram of a computing device **44** suitable for implementing any of the computing devices illustrated herein, such as the quantum computing system **12-1**, quantum computing system **12-2**, or the classical computing device **38**. The computing device **44** may comprise any classical computing device or quantum computing system capable of including firmware, hardware, and/or executing software instructions to implement the functionality described herein. The computing device **44** includes a processor device **46**, a memory **48**, and a system bus **50**. The system bus **50** provides an interface for system components including, but not limited to, the memory **48** and the processor device **46**. The processor device **46** can be any commercially available or proprietary processor.
(44) The system bus **50** may be any of several types of bus structures that may further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and/or a local bus using any of a variety of commercially available bus architectures. The memory **48** may include non-volatile memory **52** (e.g., read-only memory (ROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), etc.), and volatile memory **54** (e.g., random-access memory (RAM)). A basic input/output system (BIOS) **56** may be stored in the non-volatile memory **52** and can include the basic routines that help to transfer information between elements within the computing device **44**. The volatile memory **54** may also include a high-speed RAM, such as static RAM, for caching data.
(45) The computing device **44** may further include or be coupled to a non-transitory computer-readable storage medium such as a storage device **58**, which may comprise, for example, an internal or external hard disk drive (HDD) (e.g., enhanced integrated drive electronics (EIDE) or serial advanced technology attachment (SATA)), HDD (e.g., EIDE or SATA) for storage, flash memory, or the like.
(46) A number of modules can be stored in the storage device **58** and in the volatile memory **54**, including an operating system and one or more program modules, such as the MFA service **28-1** or the MFA service **28-2**, which may implement the functionality described herein in whole or in part.

All or a portion of the examples may be implemented as a computer program product **60** stored on a transitory or non-transitory computer-usable or computer-readable storage medium, such as the storage device **58**, which includes complex programming instructions, such as complex computer-readable program code, to cause the processor device **46** to carry out the steps described herein. Thus, the computer-readable program code can comprise software instructions for implementing the functionality of the examples described herein when executed on the processor device **46**. The processor device **46**, in conjunction with the MFA services **28-1** or **28-2** in the volatile memory **54**, may serve as a controller, or control system, for the computing device **44** that is to implement the functionality described herein.
(47) An operator may also be able to enter one or more configuration commands through a keyboard (not illustrated), a pointing device such as a mouse (not illustrated), or a touch-sensitive surface such as a display device. Such input devices may be connected to the processor device **46** through an input device interface **62** that is coupled to the system bus **50** but can be connected by other interfaces such as a parallel port, an Institute of Electrical and Electronic Engineers (IEEE) 1394 serial port, a Universal Serial Bus (USB) port, an IR interface, and the like. The computing device **44** may also include a communications interface **64** suitable for communicating with a network as appropriate or desired.
(48) Individuals will recognize improvements and modifications to the preferred examples of the disclosure. All such improvements and modifications are considered within the scope of the concepts disclosed herein and the claims that follow.

Claims

1. A method comprising: receiving, by a computing device comprising a processor device, an access request to allow a requestor process to access a protected resource; sending, by the computing device to the requestor process, a qubit authentication state that identifies a desired state of a qubit; determining, by the computing device, that the requestor process has manipulated physical characteristics of the qubit, such that the physical characteristics of the qubit match the qubit authentication state; and based at least in part on determining that the requestor process has manipulated the physical characteristics of the qubit to match the qubit authentication state, granting, by the computing device to the requestor process, access to the protected resource.
2. The method of claim 1 further comprising: providing, by the computing device to the requestor process, a qubit identifier that identifies the qubit on a quantum computing system which the requestor process is to manipulate to match the qubit authentication state.
3. The method of claim 1 further comprising: prior to sending the qubit authentication state that identifies the desired state of the qubit to the requestor process, sending, by the computing device to a first quantum computing system, a request for a quantum authentication; and receiving, by the computing device from the first quantum computing system, the qubit authentication state that identifies the desired state of the qubit.
4. The method of claim 3 wherein the computing device comprises a second quantum computing system.
5. The method of claim 4 further comprising: receiving, by the second quantum computing system, a message indicating that the requestor process has manipulated the physical characteristics of the qubit on the second quantum computing system to match the qubit authentication state; and in response to the message, causing the second quantum computing system to teleport, using quantum teleportation, the physical characteristics of the qubit on the second quantum computing system to the first quantum computing system.
6. The method of claim 5 further comprising: receiving, by the first quantum computing system, the request for the quantum authentication; generating the qubit authentication state; sending, by the first quantum computing system to the second quantum computing system, the qubit authentication

state; determining, by the first quantum computing system, that a qubit implemented by the first quantum computing system has been altered; determining that the qubit implemented by the first quantum computing system matches the qubit authentication state; and in response to determining that the qubit implemented by the first quantum computing system matches the qubit authentication state, sending, by the first quantum computing system to the second quantum computing system, a successful quantum authentication message indicating a successful quantum authentication by the requestor process.

7. The method of claim 1 further comprising: sending, by the computing device to the requestor process, a request for a classical authentication; receiving, by the computing device from the requestor process, authentication information; determining, by the computing device, that the authentication information is valid; and wherein granting, by the computing device to the requestor process, the access to the protected resource further comprises: based on determining that the requestor process has manipulated the physical characteristics of the qubit to match the qubit authentication state and on the authentication information, granting, by the computing device to the requestor process, the access to the protected resource.

8. The method of claim 1 wherein the qubit authentication state comprises a specified spin state and a specified polarization state.

9. The method of claim 1 wherein the qubit authentication state comprises a textual identifier.

10. The method of claim 1 wherein the protected resource comprises one of a file and an online service.

11. A method comprising: receiving, by a computing device comprising a processor device, an access request to allow a requestor process to access a protected resource; sending, by the computing device to a first quantum computing system, a request for a quantum authentication; receiving, by the computing device from the first quantum computing system, a qubit authentication state that identifies a desired state of a qubit; sending, by the computing device to the requestor process, the qubit authentication state; receiving, by the computing device from the first quantum computing system, a successful quantum authentication message indicating a successful quantum authentication by the requestor process, wherein the successful quantum authentication indicates that the requestor process has manipulated physical characteristics of the qubit, such that the physical characteristics of the qubit match the qubit authentication state; and based at least in part on the successful quantum authentication message, granting, by the computing device to the requestor process, access to the protected resource.

12. The method of claim 11 further comprising: sending, by the computing device to the requestor process, a request for a classical authentication; receiving, by the computing device from the requestor process, authentication information; determining, by the computing device, that the authentication information is valid; and wherein granting, by the computing device to the requestor process, the access to the protected resource further comprises: based on the successful quantum authentication message and on the authentication information, granting, by the computing device to the requestor process, the access to the protected resource.

13. The method of claim 11 wherein the qubit authentication state comprises a specified spin state and a specified polarization state.

14. A computing device, comprising: a memory; and a processor device coupled to the memory to: receive an access request to allow a requestor process to access a protected resource; send, to the requestor process, a qubit authentication state that identifies a desired state of a qubit; determine that the requestor process has manipulated physical characteristics of the qubit, such that the physical characteristics of the qubit match the qubit authentication state; and based at least in part on determining that the requestor process has manipulated the physical characteristics of the qubit to match the qubit authentication state, grant, to the requestor process, access to the protected resource.

15. The computing device of claim 14, wherein the processor device is further to provide, to the

requestor process, a qubit identifier that identifies the qubit on a quantum computing system which the requestor process is to manipulate to match the qubit authentication state.

16. The computing device of claim 14, wherein the processor device is further to: prior to sending the qubit authentication state that identifies the desired state of the qubit to the requestor process, send, to a first quantum computing system, a request for a quantum authentication; and receive, from the first quantum computing system, the qubit authentication state that identifies the desired state of the qubit.

17. The computing device of claim 16 wherein the computing device comprises a second quantum computing system.

18. The computing device of claim 14 wherein the qubit authentication state comprises a specified spin state and a specified polarization state.

19. The computing device of claim 14 wherein the qubit authentication state comprises a textual identifier.

20. The computing device of claim 14 wherein the protected resource comprises one of a file and an online service.